Crack Growth Prognostics with Particle Filtering Technique

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Abstract

A common situation in industrial plants or engineering systems is having structures and components operating under extremely harsh conditions in terms of loads, pressures, and temperatures. Moreover, especially in the nuclear field, aggressive chemical agents and nuclear radiation can play an additional role in determining a premature and irreversible degradation state for these items, especially when a combination of the degradation sources is present. For example, neutron flux (and its time-integrated counterpart, i.e. the neutron fluence) play an important role in inducing inelastic phenomena that reduce the fatigue resistance of non-irradiated objects. Procedures able to predict the future degradation state of the structures and components are used to assure a safe, reliable, and profitable operation of the complex systems which are built as a combination of elementary items such as pipes, valves, and beams. In this investigation, the main degradation mechanism of interest is assumed to be of mechanical nature, coinciding with 1D crack propagation. Then, this paper investigates the application of a model-based approach to predict the Remaining Useful Life (RUL) of a plate under loading conditions. For the solution of this prognostic problem, a Particle Filter (PF) embedding Sampling Importance Resampling (SIR) algorithm is implemented. Combining the state-space evolution of the nonlinear dynamical system with the estimation of the Probability Density Function (PDF) of the evolving state using Bayes' rule, the RUL of the loaded plate is estimated, accounting for uncertainties. A confrontation of the results with the expected outcomes is possible, thanks to the physics-based model adopted to describe the crack propagation in the plate and solving the ODE that characterizes the damage progression model. In this work, some features of the crack growth model are highlighted, and then the development of the particle filtering technique follows. Finally, the prediction abilities and features of the method are discussed.

Keywords: Crack growth, RUL estimation, Particle Filters with SIR, Bayesian framework.

1 Introduction

The relevance of a crack growth investigation method for prognostics in the nuclear industry is directly evident when considering the fundamental Deterministic Safety Assessment (DSA) and Probabilistic Safety Assessment (PSA). For example, Ref. [3] discusses how the Steam Generator (SG) is one of the most important safety components in a Nuclear Power Plant (NPP). Accidents caused by Stress Corrosion Cracking (SCC), a phenomenon in which mechanical and chemical agents degrade the interested structure, have been reported. Consequently, Steam Generator Tube Rupture (SGTR) is considered one of the initiating events when performing a DSA or/and a PSA. Then, a PSA technique such as the Failure Modes, Effects and Criticality Analysis (FMECA) can highlight the need for a detailed understanding of the dynamics with which the physical phenomenon of crack propagation happens.

The current work deals with a branch of maintenance management called prognostics. It aims at exploiting the specific information on the degradation model available for the specific degradation process to implement a model-based method to infer the life evolution of the state of the selected component, using a sequence of noisy measurements describing its evolving state. These measurements somehow reflect the operational history and the environmental conditions in which the system operates, thus characterizing the degradation process in a unique way.

Given the current problem of crack propagation, PF demonstrates its effectiveness as a support decision tool for a condition-based maintenance strategy implementation, defining a suitable probabilistic framework for evaluating the RUL of the investigated component. Particle Filters are sequential Monte Carlo methods: although being suboptimal, they show advantage in applications for nonlinear systems with non-Gaussian noise terms (such as fatigue crack growth), where optimal solutions are not feasible.

2 Data and Model Presentation

The Paris-Erdogan model for crack propagation is assumed:

$$\frac{da}{dN} = C\,(\Delta\sigma\,\sqrt{\pi a})^m. \tag{1}$$

In the previous expression, a [m] is the crack length; C and m are constants that depend on the material constituting the plate and the depending on the environment in which the object operates. The range of the stress intensity factor is written as $\Delta k = \Delta\sigma\,\sqrt{\pi a}$, where $\Delta\sigma$ [MPa] is the stress range. The independent variable, which plays the role of a temporal variable, is the number of cycles N [−].

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The state propagation departs from Equation (1), supposing a sufficiently small $dN \approx \Delta N$, and with $da \approx \Delta a = a_{k+1} - a_k$. The following recurrence relation representing a nonlinear Markov process with hidden state a_k is then obtained:

$$a_{k+1} = a_k + C_k \left(\Delta\sigma \sqrt{\pi a_k}\right)^{m_k} \Delta N. \tag{2}$$

Notice that in the current work the model parameters m and C are unknown, so the procedure turns out to be a model-based joint state-parameter estimation, in which both the state (distribution) a_k and the prior and posterior parameters distributions are approximated as a set of discrete and weighted samples.

For the current problem, the available dataset consists in a sequence of 25 measurements of the degradation state, registered every $\Delta N = 50$ loading cycles. Other data and initial distributions are listed in Table 1.

Table 1: Data for the crack growth problem.

Quantity	**Symbol**	**Value**
True value of m	m_{true} [−]	3.8
True value of C	C_{true} [−]	1.5×10^{-10}
Stress range	$\Delta\sigma$ [MPa]	78
Failure threshold	a_{FAIL} [m]	0.0463
Initial dist. (a)	a_0 [m]	$\sim \mathcal{N}(0.01;\ 5 \times 10^{-4})$
Initial dist. (m)	m_0 [−]	$\sim \mathcal{N}(4;\ 0.04)$
Initial dist. (C)	$\ln(C_0)$ [−]	$\sim \mathcal{N}(-22.33;\ 1.2544)$
Meas. uncertainty	σ_ν [m]	$\sim \mathcal{N}(0.001;\ 0)$

Consider that due to the very small values for C and to avoid numerical problems, $\ln(C)$ is used in place of C, thus considering $\exp(\ln C)$ in Equation (2). Taking into account that the load cycles are in total 25, ranging from 0 to 1200 with step $\Delta N = 50$, a proper ODE solver available in Python has been implemented, so that in the single plot of Figure 1 a concurrent representation of the exact solution and the measured values of crack growth is given.

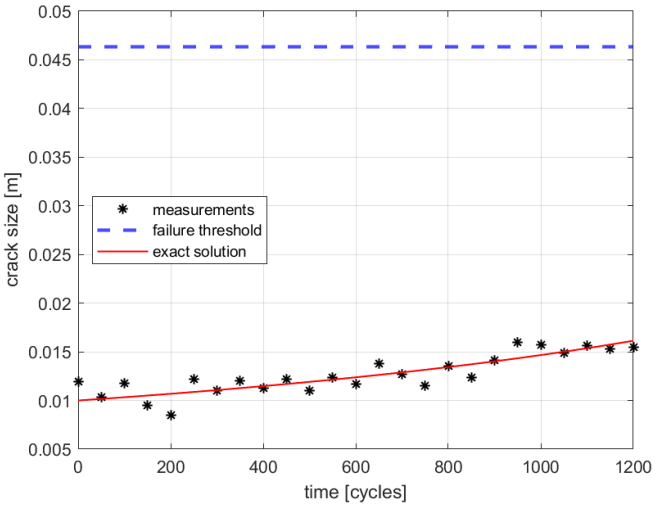


Figure 1: Initial data for crack growth estimation.

3 Particle Filtering Method Fundamentals

A detailed explanation of the properties of Particle Filters for Bayesian state-parameter tracking is presented in [1]. Hereafter, only the fundamental equations are recalled and explained. The PF is based on the state transition function f and the measurement equation h, evaluated at each timestep k:

$$\begin{cases} x_{k+1} = f(x_k,\, \theta_k,\, \omega_k); \\ z_k = h(x_k,\, \nu_k). \end{cases} \tag{3}$$

In these equations, x_k is the damage state, θ_k is a vector of model parameters, z_k are measurement data and ω_k, ν_k are the noise terms affecting the state and the measurements, respectively. Notice that Equation (2) is just a specific case of Equation (3a), in which the presence of process noise is ignored and handled through the uncertainty in the model parameters. Concerning the notation denoting the damage state, the symbol a_k will be adopted in Section (4), sticking to the notation adopted in Equation (2).

PF is a recursive filter that predicts the state at timeframe k using the information at $k-1$ as a prior, and next updates the predicted state relying on the measurement data, using the concept of likelihood. This allows the posterior distribution of the state to be obtained and interpreted as the prior for step $k+1$ of the sequential method. Within a Bayesian framework, the posterior distribution, represented by $p(x_k(\theta_k) \,|\, z_{0:k})$, is determined alternating a prediction stage and an update stage, as described below.

Introducing $\mathcal{L}(z_k \,|\, x_k(\theta_k))$ as the likelihood (which represents the PDF of z_k conditional on the given $x_k(\theta_k)$) and $p(x_{k-1}(\theta_{k-1}))$ as the prior PDF, the use of Bayes' theorem allows the evaluation of the following PDF:

$$p(x_k(\theta_k) \,|\, z_{0:k}) \propto \mathcal{L}(z_k \,|\, x_k(\theta_k)) \cdot p(x_{k-1}(\theta_{k-1}))\,. \tag{4}$$

4 State Tracking Using Particle Filter

With PF, the state distribution is approximated by a set of N_p discrete weighted samples, called particles

$$\{a_k^i,\, w_k^i\}_{i=1}^{N_p}, \tag{5}$$

where a_k^i, i.e. x_k^i, denotes the state estimate constituted by particle i, and w_k^i denotes the corresponding weight of the particle i. From now on, the probability density

functions are described by a collection of particles:

$$p(a_k \mid z_{0:k}) \approx \sum_{i=1}^{N_p} w_k^i \, \delta\left(a_k - a_k^i\right). \tag{6}$$

For the crack growth degradation problem under analysis, in Ref. [2] a lognormal likelihood is suggested:

$$\mathcal{L}\left(z_k \mid a_k^i, m_k^i, C_k^i\right) = \frac{1}{\sqrt{2\pi}\, z_k\, \zeta_k^i} \exp\left[-\left(\frac{\ln z_k - \lambda_k^i}{\sqrt{2}\zeta_k^i}\right)^2\right], \tag{7}$$

where the index i addresses the specific particle and ζ_k^i, λ_k^i have the following expressions:

$$\begin{cases} \zeta_k^i = \sqrt{\ln\left(1 + \left(\frac{\sigma}{a_k^i}\right)^2\right)}, \\ \\ \lambda_k^i = \ln\left(a_k^i\right) - \frac{1}{2}\left(\zeta_k^i\right)^2. \end{cases} \tag{8}$$

As anticipated in Section 3, the iterative procedure is divided into two stages. First, the parameters and the state is initialized performing a random extraction random of values from the distributions listed in Table 1, having fixed the number of particles $N_p = 5000$. Statistical indicators are computed for each time-step, referring to the whole population of particles at time-frame k and allowing for the tracking of the state and of the parameters. At each time-frame, the state is propagated using Equation (2) for all the N_p particles. Next, N_p values of likelihood are computed using Equations (7–8) and the measurements available at time-frame k.

After that, weight update and normalization follow for each one of the particles, using:

$$w_k^i = \frac{w_{k-1}^i \, \mathcal{L}\left(z_k \mid a_k^i, m_k^i, C_k^i\right)}{\sum_j \mathcal{L}\left(z_k \,\middle|\, a_k^j, m_k^j, C_k^j\right)} \tag{9}$$

The SIR algorithm is implemented to avoid the degeneracy of the weights w_k^i of the particles, using the inverse Cumulative Distribution Function (CDF) method. To such an extent, the CDF of the particles is constructed using the weights (i.e., the likelihood) and then a random value $r \sim \mathcal{U}(0;1)$ is drawn to identify a specific particle in this CDF, which of course takes values in the interval $(0;1)$. The index for which the CDF is greater than r is found and substitution of that particle occurs in the population of the states and parameters using such index. Finally, weights are reassigned as $1/N_p$. In total, this procedure runs $l = 25$ times.

5 RUL Prediction

The prediction starts at timestep $k = l + 1$: particle weights and parameters are kept fixed at their last update

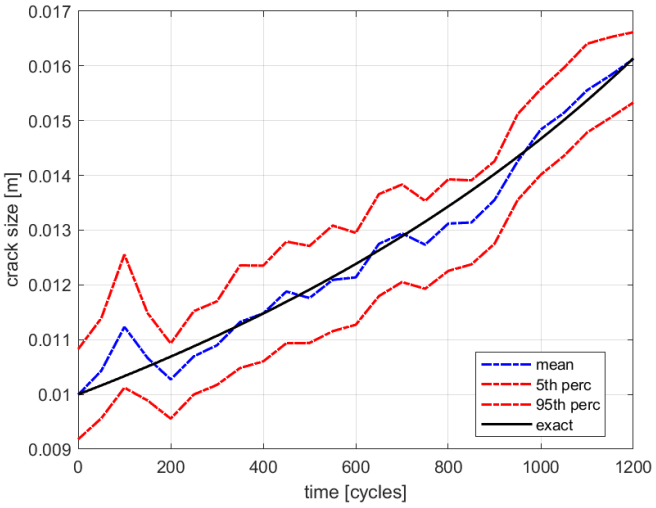


Figure 2: State tracking with Particle Filter.

and the state of all the particles is propagated up to the failure threshold. As explained in Refs. [2] - [3], the correct way to propagate the state is by means of Equation (2), but accounting for uncertainty during the evolution.

For each particle, the End-Of-Life (EOL) variable is introduced and incremented cycle after cycle as the state of the corresponding particle remains below the fixed threshold during the simulation. The threshold, of course, reflects the a-priori fixed allowable damage state, which in this case is the crack length. Once the EOL is updated and the state propagated, the value of the state is resampled using a lognormal distribution, whose mean and variance are dependent on the previous value of the state and on the uncertainty which characterizes the measurements, i.e. $\sigma_{\nu_{\text{measure}}}$ [m]. By doing so, a population of end-of-life time-steps is obtained, allowing for the calculation of a population of RULs. In fact, in terms of load cycles, it results:

$$\text{RUL}(i) = t_{\text{EOL}}(i) - l\,. \tag{10}$$

Then, it is possible to represent the RUL distribution and calculate statistical indicators. Figure (3) gives a direct representation of the statistical indicators concerning the RUL, while Figure (4) gives a representation of the evolution in time of the damage state, which for this problem is the crack length, quoted in (m).

6 Results for Prognostics

The purpose of prognostics is to use the information available on the component to predict and manage future failures. Once the RUL of an item is predicted, decisions regarding maintenance management (e.g., repair/substitute the item or not) must be made. Thus, it is fundamental to have measures that can convey the confidence level of the RUL prediction and its degree of trust.

First, the true RUL of the item has been calculated using

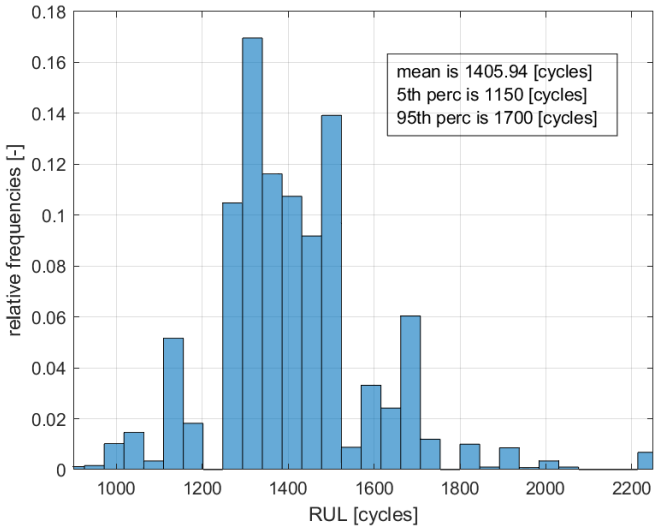


Figure 3: RUL histogram.

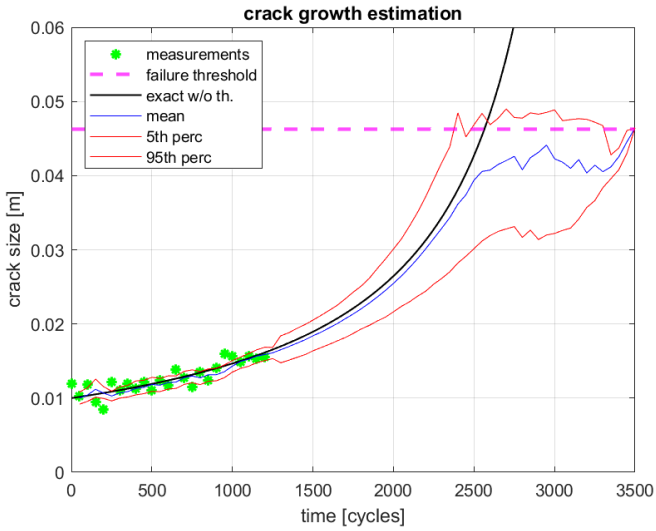


Figure 4: Crack growth estimation during prognostics.

the ODE solver: $\text{RUL}_{\text{true}} = 1350$ (cycles). Having this information, it was possible to calculate error indicators such as the mean relative absolute error and the root mean squared error: MRAE $= 0.104$, RMSE $= 195.095$. In addition, a measure of the dispersion of RULs within a $\pm\hat{\sigma}$ band with respect to the mean value was computed, obtaining the coverage measure: cov $= 73.78\%$. This indicator tells that the signal is concentrated, thus the statistical indicators are believed sufficiently robust and indicative.

To gain insight into the evolution of the state, Figure (4) is presented: the mean value of the state, the 5th and 95th percentiles are able to approximate accurately the exact solution up to a point (cycles $\cong 2500$) from which the population of particles starts being depleted, due to the fact that the crack depth x_k^i reaches the threshold and the corresponding particle trajectory ends. For larger times, the estimates for state and percentiles are less robust because the number of trajectories contributing to the estimates is greatly reduced, due to the above mentioned depletion. For the previous reasons, it is possible to say that the results of the state analysis can be seen as quite accurate and with an acceptable level of confidence up to a time-frame equal circa to twice the initial service time ($2500 \approx 2 \times 1200$) of the item. However, also the jagged part of the estimates seems satisfactory.

7 Conclusion

In this work, Particle Filtering technique has been applied for the prediction of the RUL of a loaded plate, a quite common situation in industrial systems. The technique proved its ability to detect the degradation phenomenon and estimate its state, also accounting for uncertainties, as the error indicators testify. Further developments may involve new predictions and confrontations if measurements for $t > 1200$ cycles are provided.

Acknowledgments: The author wishes to thank D. An *et al.* for having published the paper [2]: *"A tutorial for Model-based Prognostics Algorithms based on Matlab Code"*, in which all the data used in this work have been found.

References

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